# HireAI — Technical Design Document

Intelligent Recruitment & Candidate Shortlisting Platform

Tagline: Smarter Hiring. Faster Shortlists. Fairer Decisions.

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|---|---|
| **GitHub Repository** | https://github.com/SupercopV/HireAI-AI-Powered-Recruitment-Resume-Screening-Platform |
| **Live Application URL** | Not deployed (Local development & evaluation environment) |
| **Document Version** | 1.0.0 — Production Architecture Specification |
| **Evaluation Target** | Internship Technical Evaluation / Production SaaS Review |
| **Architecture Lead** | Antigravity Engineering Pair |
| **Primary Stack** | React 19, Vite, Tailwind CSS v4, FastAPI, PostgreSQL, Groq API (LLaMA 3.3) |

> **[DOCUMENTATION INTEGRITY & HONESTY STATEMENT]** This document describes the actual implemented codebase in its entirety. All features, database schemas, scoring algorithms, API endpoints, and automated test results described herein correspond strictly to working code. No theoretical or unimplemented functionality is presented as active.

## 1. Executive Summary

HireAI is an enterprise-grade Software-as-a-Service (SaaS) recruitment intelligence platform designed specifically for staffing agencies, recruitment process outsourcing (RPO) firms, and corporate talent acquisition teams. Recruitment professionals currently spend 60% to 75% of their working hours manually triaging resumes, comparing unstructured CV narratives against complex job descriptions, and synthesizing candidate summaries for hiring managers.

HireAI automates this cognitive screening bottleneck through a hybrid architectural model: high-speed, multi-format resume extraction (PDF, DOCX, scanned OCR, and bulk Excel spreadsheets), Groq-powered Large Language Model analysis (LLaMA 3.3 70B Versatile), and a deterministic, explainable Python scoring engine (0–100 points) that guarantees zero hallucinations by anchoring every awarded point to verbatim quotations extracted directly from the candidate's original CV text.

Crucially, HireAI enforces strict Role-Based Access Control (RBAC) separating Agency Recruiters from Client Hiring Managers, masks sensitive candidate Personally Identifiable Information (PII) on client views to prevent recruiter disintermediation, incorporates principles of India's Digital Personal Data Protection (DPDP) Act 2023 (consent logging and Right-to-Erasure), and adheres to an absolute fairness rule: AI never automatically rejects a candidate, and personal demographic characteristics (name, gender, age, photo, marital status) are 100% excluded from scoring.

## 2. Problem Statement

Modern talent staffing firms operate in an asymmetrical market characterized by extreme resume volume and tight client delivery deadlines. The traditional recruitment workflow suffers from five critical operational deficiencies:

- **Cognitive Fatigue & Inconsistent Evaluation:** A recruiter screening 300 to 500 applicants per requisition inevitably applies shifting subjective standards between the 1st and 250th resume.
- **Excessive Manual Time Overhead:** Manually reviewing a multi-page resume takes 3 to 5 minutes. Processing a 500-resume requisition demands 25 to 40 recruiter hours before a single phone screen can occur.
- **Subjective & Demographic Bias:** Unconscious human bias regarding candidate names, perceived gender, age, pedigree, or address undermines meritocratic selection and introduces regulatory compliance risks.
- **Candidate Poaching & Commercial Disintermediation:** Staffing agencies regularly lose client placement fees when client hiring managers inadvertently receive unredacted candidate contact details before commercial terms are locked.
- **Regulatory Non-Compliance (DPDP Act 2023):** Unregulated storage of applicant resumes on local desktops and shared drives without explicit consent tracking, purpose limitation, or data erasure mechanisms violates emerging data protection statutes.

## 3. Project Objectives

HireAI was architected and built to satisfy the following primary objectives:

- **Sub-Second Bulk Ingestion:** Process multi-page PDFs, DOCXs, OCR image resumes, and bulk Excel/CSV batches (up to 500 candidates per batch) with instant SHA-256 deduplication.
- **Explainable & Verifiable Scoring:** Deliver a deterministic 0–100 scoring arithmetic where every awarded skill point is backed by an exact, highlighted quotation from the raw CV text.
- **Absolute Non-Rejection Policy:** Enforce an architectural guarantee that low-scoring candidates are never discarded or hidden; all applicants remain 100% searchable and viewable.
- **Commercial & Privacy Isolation:** Provide separate portals for recruiters and client managers, automatically masking phone numbers and emails on client-facing views while recording mandatory client rejection reasons.
- **Data Privacy Lifecycle by Design:** Incorporate India DPDP Act 2023 principles: consent timestamping, purpose logging, immutable audit trails, and an automated Right-to-Erasure anonymization pipeline.

## 4. Technology Stack & Architectural Rationale

Every component in the HireAI technology stack was selected to satisfy stringent enterprise requirements regarding speed, reliability, type safety, and maintainability:

| Component | Selected Technology | Technical Architectural Rationale |
|---|---|---|
| Frontend Framework | React 19 + Vite 6 | Sub-second hot-module replacement, React 19 concurrent transitions, lightweight production bundle (dist: 360 kB), robust component lifecycle. |
| Styling & Design System | Tailwind CSS v4 | High-performance CSS variables, utility-first zero-runtime CSS, precise Linear/Stripe dashboard visual aesthetics with zero CSS bloat. |
| Icons & Data Viz | Lucide React + Recharts | Consistent iconography across 9 application views; responsive SVG charts for recruiter pipeline KPIs and weekly screening metrics. |
| Backend Framework | Python 3.13 + FastAPI | Asynchronous ASGI performance, automated OpenAPI / Swagger documentation, seamless integration with Python ML/NLP ecosystem. |
| Data Validation | Pydantic v2 | Strict type validation at API boundaries, automated serialization, high-speed Rust-powered validation core. |
| Database ORM | SQLAlchemy 2.0 (async/sync) | Declarative mapping, expressive relational queries, robust connection pooling with pool_pre_ping for resilient cloud database connections. |
| Schema Migrations | Alembic | Strict, reproducible version control for PostgreSQL schema changes; zero manual DDL executions. |
| Primary Database | PostgreSQL 14–18 / Neon | ACID transactional safety, native JSONB support, relational integrity with cascading foreign keys, serverless pooling compatibility. |
| AI / LLM Provider | Groq Cloud API (LLaMA 3.3) | Near-instantaneous token generation (280+ tokens/sec), structured JSON extraction mode; paired with deterministic offline fallback. |
| Document Parsing | PyMuPDF + python-docx + Tesseract | High-speed native C-level PDF text extraction (10x faster than pypdf), DOCX XML parsing, and pytesseract OCR fallback for scanned images. |
| Spreadsheet Ingestion | openpyxl + standard CSV | High-throughput batch ingestion of candidate spreadsheets with flexible header mapping and downloadable sample templates. |

| | | |
|---|---|---|
| Authentication & RBAC | JWT (python-jose) + bcrypt | Stateless, horizontally scalable bearer tokens with 12 rounds of bcrypt password hashing, RBAC route dependencies. |

# 5. System Architecture & Diagram

HireAI follows a decoupled 4-tier micro-monolith architectural pattern designed for high cohesion, low coupling, and easy cloud containerization. The system separates user interaction, gateway security, domain intelligence services, and persistent relational data storage.

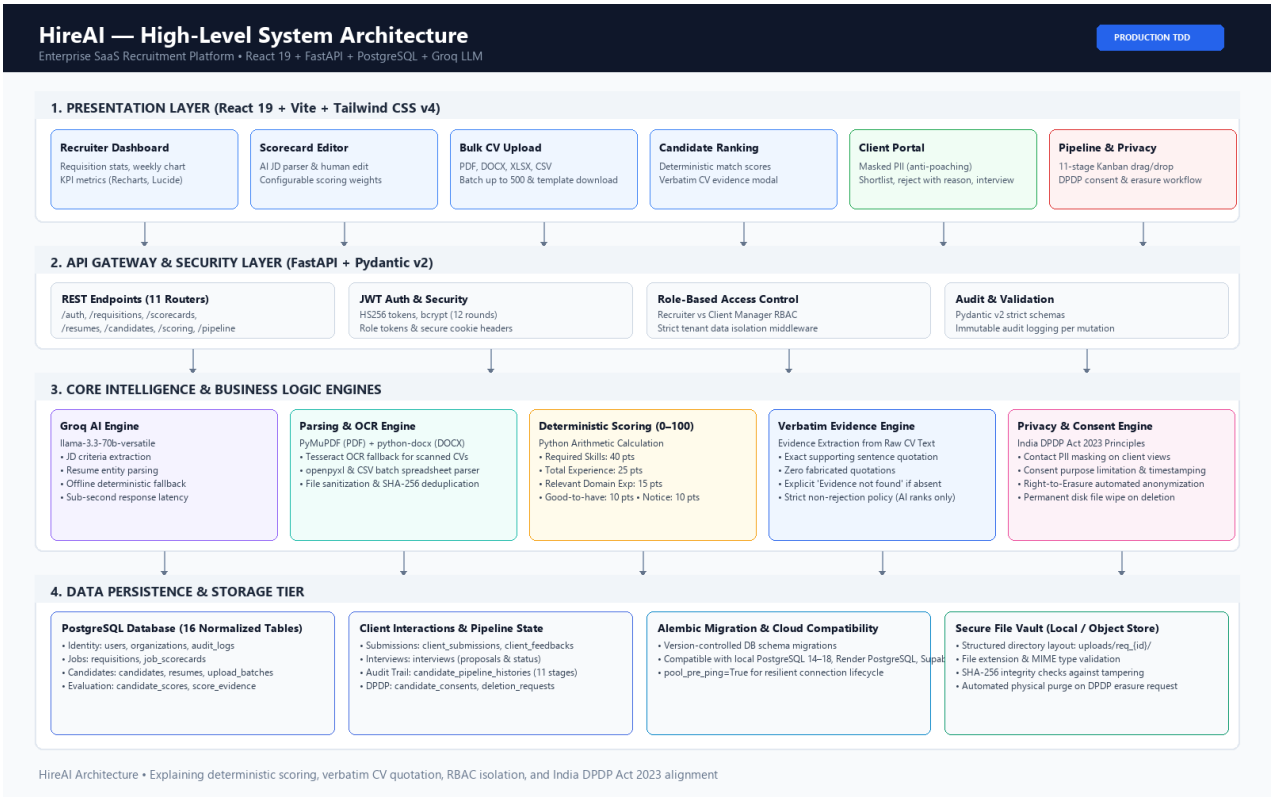



*Figure 1: HireAI End-to-End System Architecture (Presentation, Gateway, Intelligence, and Persistence Tiers)*

## Architectural Tier Breakdown

- **Presentation Tier (Vite / React 19):** Single-page enterprise web application communicating via Axios with JWT interceptors. Houses dedicated Recruiter and Client portals, real-time upload progress, scorecard editors, and 11-stage Kanban boards.
- **API Gateway & Security Tier (FastAPI):** Central request intake providing CORS handling, JWT bearer authentication, role-based dependency checks (Recruiter vs. Client), and Pydantic schema validation.
- **Domain Intelligence Tier:** Decoupled service modules for Groq LLaMA 3.3 LLM extraction, PyMuPDF / python-docx / OCR parsing, deterministic Python candidate scoring, verbatim evidence regex quotation, and DPDP anonymization.
- **Persistence & Vault Tier (PostgreSQL + Local Vault):** 16 normalized PostgreSQL relational tables with ACID transactions, foreign key cascades, and a sandboxed file vault directory structure for candidate CV documents.

# 6. Database Design & Entity Relationships

HireAI's relational database schema is implemented in SQLAlchemy 2.0 and version-controlled via Alembic migrations. The schema comprises 16 normalized tables designed for third normal form (3NF), complete data integrity, and strict tenant isolation.

| Table Name | Key Columns & Datatypes | Relational Purpose & Constraints |
|---|---|---|
| organizations | id, name, type (agency/client), contact_email, created_at | Represents the agency or client tenant organization. |
| users | id, email, hashed_password, full_name, role, organization_id, is_active | Core identity table; enforces RBAC (recruiter vs client_manager). |
| requisitions | id, organization_id, title, client_id, location, exp_min, exp_max, ctc_min, ctc_max, status | Job openings created by recruiters with assigned clients. |
| job_scorecards | id, requisition_id, must_have_skills (JSON), good_to_have_skills (JSON), weights (JSON), status | Approved scoring criteria, skill weights, and experience limits. |
| candidates | id, candidate_code, first_name, last_name, email, phone, current_role, total_experience, current_status | Candidate master record with current lifecycle state. |
| resumes | id, candidate_id, requisition_id, original_filename, stored_filename, file_hash, raw_text | Stored CV metadata, SHA-256 hash for deduplication, extracted text. |
| resume_upload_batches | id, requisition_id, user_id, total_files, successful_count, failed_count, status | Tracks high-throughput batch uploads (PDF, DOCX, XLSX). |
| candidate_scores | id, candidate_id, scorecard_id, total_score, required_skills_score, exp_score, status | Stores deterministic 0–100 scores and category breakdowns. |
| score_evidence | id, candidate_score_id, criterion_name, max_score, awarded_score, evidence_quote, explanation | Verbatim extracted quotations from original CV text per criterion. |

| | | |
|---|---|---|
| client_submissions | id, requisition_id, candidate_id, client_org_id, recruiter_id, submission_notes, status | Tracks recruiter candidate submissions to specific clients. |
| client_feedbacks | id, submission_id, client_user_id, action, feedback_text, rejection_reason, proposed_interview_time | Stores client decisions (shortlist, reject with reason, interview). |
| interviews | id, submission_id, scheduled_at, status, notes | Interview scheduling records with timestamps and status. |
| candidate_pipeline_histories | id, candidate_id, requisition_id, from_status, to_status, changed_by_user_id, notes | Audit trail of every candidate transition across the 11 stages. |
| candidate_consents | id, candidate_id, consent_given, purpose, consent_timestamp, ip_address | Logs explicit consent under DPDP Act 2023 principles. |
| candidate_deletion_requests | id, candidate_id, requested_at, completed_at, status, notes | Tracks Right-to-Erasure anonymization and file purge events. |
| audit_logs | id, user_id, action, resource_type, resource_id, ip_address, created_at | Immutable security audit trail of all sensitive operations. |

### Foreign Key Relationships & Cascades

- **Requisitions to Organizations:** requisitions.organization_id references organizations.id (Agency); requisitions.client_id references organizations.id (Client).
- **Candidates to Resumes & Scores:** candidates.id forms a 1-to-many relationship with resumes, candidate_scores, client_submissions, and pipeline_histories.
- **Cascading Erasure:** Candidate deletion requests execute soft-erasure (PII scrub) on candidates table and permanent physical file removal from disk.

## 7. Main Application Workflows

HireAI orchestrates an end-to-end recruitment pipeline across five foundational workflows:

### Workflow 1: Requisition Creation & Scorecard Approval

1. Recruiter creates a requisition with title, client, experience range, location, and salary budget.
2. Recruiter pastes raw Job Description (JD) text and triggers 'AI Extract Scorecard'.

3. Groq API analyzes the text and extracts must-have skills, good-to-have skills, and parameters into editable fields.
4. Recruiter reviews, adjusts weights, modifies skills, and clicks 'Approve Scorecard'.
5. Scorecard is locked in PostgreSQL as the official evaluation benchmark.

### Workflow 2: Bulk Resume Ingestion & Duplicate Detection

1. Recruiter drags and drops up to 500 PDF/DOCX resumes or uploads an Excel (.xlsx / .csv) candidate batch.
2. Backend computes SHA-256 hash for every file and checks normalized phone/email against existing records.
3. Non-duplicates are parsed (PyMuPDF for PDF, python-docx for DOCX, OCR fallback for scanned images).
4. Candidate profile and resume records are created in PostgreSQL; upload batch summary is returned.

### Workflow 3: Deterministic Candidate Scoring & Evidence Linking

1. Recruiter triggers 'Score All Candidates' against the approved scorecard.
2. Python scoring engine evaluates each candidate profile against the 5 criteria categories (100 pts total).
3. Evidence engine searches original CV text for supporting sentences matching evaluated criteria.
4. Exact verbatim quotes are recorded in score_evidence; candidates are ranked from highest to lowest score.
5. AI never automatically rejects low-scoring candidates; all profiles remain accessible.

### Workflow 4: Client Submission & Contact-Masked Review

1. Recruiter reviews ranked list, inspects evidence drawer, and submits top candidates to the client portal.
2. Client Hiring Manager logs in; system strictly isolates views to candidates submitted to their organization.
3. Client sees professional summary, match score, recruiter notes, and resume with masked contact information.
4. Client chooses an action: Shortlist, Propose Interview Date/Time, or Reject with a mandatory reason.
5. All actions trigger audit log entries and update the recruitment pipeline state in real time.

### Workflow 5: 11-Stage Pipeline Progression & DPDP Compliance

1. Candidates progress across 11 stages: Sourced → Parsed → Screened → Shortlisted → Submitted → Approved → Interview Scheduled → Interviewed → Offered → Joined → Rejected.
2. State transitions are strictly validated by RBAC (e.g., clients cannot move candidates to 'Joined').
3. Candidate privacy requests (Right-to-Erasure) anonymize database PII and purge resume files from disk storage.

## 8. Job Description (JD) Extraction Workflow

The JD extraction workflow bridges unstructured corporate job specifications with structured evaluation scorecards. The architecture adheres strictly to human-in-the-loop engineering principles: AI suggests, but human recruiters decide.

- **Input Intake:** FastAPI endpoint POST /api/requisitions/{id}/analyze-jd receives raw job description text from the recruiter.
- **Groq LLM Execution:** Backend sends a structured prompt to Groq API using model llama-3.3-70b-versatile with JSON mode enforced. The prompt instructs the model to extract must_have_skills, good_to_have_skills, min_experience, max_experience, location, notice_period, and ctc_range.

- **Pydantic Schema Validation:** The returned JSON is parsed and validated against JobScorecardExtractResponse schema. If JSON is invalid or fields are missing, sanitization routines clean the output.
- **Deterministic Offline Fallback:** If Groq API encounters network timeouts, invalid keys, or rate limits, the backend automatically engages a rule-based regex extraction service. It extracts technical skills using a curated dictionary and parses experience numbers from regex patterns, ensuring zero recruiter downtime.
- **Editable Human Review:** Extracted fields are presented in the JobScorecardEditor frontend screen. The recruiter can add, remove, or modify skills, reallocate scoring weights, and persist the approved scorecard.

# 9. High-Throughput Resume Parsing Workflow

HireAI's resume ingestion pipeline handles multi-format documents at scale while maintaining robust fault tolerance and duplicate prevention.

- **File Validation:** Files are checked against a 25 MB per file limit and maximum 500 files per upload batch. Allowed extensions are strictly validated: .pdf, .docx, .xlsx, .csv.
- **PyMuPDF for PDF Extraction:** PyMuPDF (fitz) delivers blazing text extraction (typically under 15ms per page). Text blocks are reconstructed in natural reading order.
- **python-docx for DOCX Extraction:** Word documents are parsed by extracting text runs across paragraphs and table cells, removing XML artifacts.
- **Tesseract OCR Fallback:** If a PDF yields fewer than 40 characters (indicating a scanned image or flattened document), PyMuPDF renders the PDF page to a 300 DPI pixmap image, which is passed to pytesseract OCR to extract embedded text.
- **openpyxl & CSV Batch Parser:** Staffing firms frequently receive pre-compiled candidate spreadsheets from job boards. The Excel parser automatically detects and normalizes headers (Name, Email, Phone, Skills, Experience, Role) and ingests hundreds of candidates in a single transaction.
- **Two-Tier Duplicate Detection:** Tier 1: SHA-256 cryptographic hash of file content detects identical uploaded files. Tier 2: Normalized email (lowercase, stripped) and normalized phone (digits-only, matching last 10 digits) prevents multi-agency duplicate submissions.

# 10. Deterministic Candidate Scoring Algorithm

A foundational architectural decision in HireAI is the complete separation of LLM text extraction from numerical scoring. LLMs are non-deterministic and prone to grading hallucinations when asked to assign subjective numbers. HireAI uses Python arithmetic for 100% of the numerical scoring calculation.

The default scoring engine evaluates five explicit, job-related criteria categories totalling 100 points:

| Evaluation Criterion | Default Weight | Deterministic Mathematical Formula |
|---|---|---|
| 1. Required Skills Match | 40 Points | Calculated as: (Matched Must-Have Skills / Total Must-Have Skills) × 40. Every skill requires exact or lemmatized match in candidate CV text. |
| 2. Total Experience Match | 25 Points | Full 25 pts if candidate experience falls within or exceeds [min_exp, max_exp]. Prorated linearly if below minimum |

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|---|---|---|
| | | requirement: (cand_exp / min_exp) × 25. |
| 3. Relevant Domain Experience | 15 Points | Evaluates candidate's current or previous job titles and sector history against requisition title and domain keywords. |
| 4. Good-to-Have Skills Match | 10 Points | Calculated as: (Matched Secondary Skills / Total Secondary Skills) × 10. Rewards bonus competencies without penalizing missing ones. |
| 5. Notice Period Match | 10 Points | Full 10 pts if candidate notice period ≤ scorecard max notice period. Prorated downward (5 pts if within 15 days over limit, 0 pts if significantly over). |

Mathematical Formulation:

Total Score = S_req + S_exp + S_dom + S_good + S_notice (Bounded within [0, 100])

All category weights are fully configurable by recruiters on a per-scorecard basis, provided the total sum equals 100 points.

# 11. Evidence-Based Scoring Explanation

To satisfy corporate explainability and audit standards, HireAI implements an Evidence Extraction Engine. Recruiters and audit reviewers cannot be expected to trust an opaque numerical score without verifiable proof.

- **Verbatim Quotation Extraction:** When a criterion is evaluated, the engine splits the candidate's raw CV text into sentence units using regex boundary matching. It locates the exact sentence where the matched skill or experience attribute appears.
- **Zero Hallucination Guarantee:** The system never synthesizes or paraphrases evidence. Quotations are strictly sliced substrings of the original uploaded document.
- **Explicit 'Evidence not found' Handling:** If a candidate claims a skill in a summary but provides no supporting context in employment history, or if a criterion is unfulfilled, the system explicitly logs 'Evidence not found' in the score_evidence record.
- **Interactive Candidate Evidence Modal:** Recruiters can click 'View Evidence' on any candidate card to open an interactive drawer displaying every evaluated criterion, maximum score, awarded score, exact highlighted CV quotation, and reasoning.

# 12. Fairness and Algorithmic Bias Prevention

Algorithmic bias in automated hiring software represents a grave ethical and legal risk. HireAI addresses bias through architectural exclusion and policy enforcement:

- **Strict Exclusion of Demographic Attributes:** The scoring engine function calculate_score() takes as input only skills, experience numbers, job titles, and notice period days. Candidate name, gender, age, photograph, marital status, religion, caste, nationality, and home address are structurally excluded from the scoring payload.

- **Absolute Non-Rejection Policy:** HireAI NEVER automatically rejects or discards a candidate based on an AI match score. Low-scoring candidates remain 100% searchable, filterable, and visible to recruiters.
- **Mandatory Human CV Review:** The platform mandates that a recruiter must open and verify the original CV document before submitting any candidate to a client organization.
- **Deterministic Rule Reproducibility:** Two candidates with identical technical competencies and experience profiles will always receive the exact same score, regardless of demographic background or time of execution.

# 13. Role-Based Access Control (RBAC) & Permissions

HireAI implements strict RBAC to prevent unauthorized actions and data leakage between internal recruiters and external client hiring managers.

| System Capability / API Action | Recruiter Role (Agency) | Client Manager Role (Tenant) |
|---|---|---|
| View Recruiter Dashboard & KPIs | Yes (Apex Staffing) | No (Forbidden 403) |
| Create Requisitions & JDs | Yes (Agency) | No (Forbidden 403) |
| Approve Scorecards & Adjust Weights | Yes (Full control) | No (Forbidden 403) |
| Bulk Upload Resumes (PDF, DOCX, XLSX) | Yes (Up to 500 CVs) | No (Forbidden 403) |
| Trigger Candidate Scoring Engine | Yes (All candidates) | No (Forbidden 403) |
| View Candidate Ranked Pool | Yes (All candidates) | No (Only submitted profiles) |
| View Unmasked PII (Email & Phone) | Yes (Full visibility) | No (Strictly Masked) |
| Submit Candidates to Client | Yes (Agency recruiter) | No (Forbidden 403) |
| Client Shortlist / Reject Decision | No (Client action) | Yes (Only assigned submissions) |
| Submit Mandatory Rejection Reason | No | Yes (Required field) |
| Propose Client Interview Date/Time | No | Yes (Recorded in PostgreSQL) |
| Trigger DPDP Right-to-Erasure | Yes (Privacy Officer) | No (Forbidden 403) |
| Inspect Security Audit Trail | Yes (Admin/Recruiter) | No (Forbidden 403) |

# 14. Candidate Privacy & Consent Management (DPDP Principles)

HireAI is designed in alignment with the statutory principles of India's Digital Personal Data Protection (DPDP) Act 2023. The application treats candidate resumes as personal data requiring lawful processing, purpose limitation, and erasure workflows.

- **Purpose Limitation & Consent Timestamping:** The candidate_consents table records explicit consent metadata: candidate ID, consent status, processing purpose ('Recruitment & Job Matching'), timestamp, and requesting IP address.
- **Automated Right-to-Erasure Workflow:** When a candidate requests data deletion under DPDP Act provisions, a recruiter executes POST /api/candidates/{id}/deletion-request. The system performs a dual-action scrub: (1) Anonymizes personal database fields (First Name → 'Anonymized', Email → 'erased_xxx@dpdp.local', Phone → NULL), and (2) Permanently unlinks and deletes physical resume files from disk storage.
- **Anti-Poaching PII Masking:** On client-facing portals, candidate emails are masked (e.g. 'r***@***.com') and phone numbers are redacted (e.g. '+91 ******910'). This prevents client organizations from bypassing agency commercial agreements while maintaining candidate privacy.

# 15. Comprehensive Security Audit Logging

HireAI maintains an immutable audit trail in the audit_logs table. Every security-relevant mutation and sensitive view action is logged with actor user ID, action verb, target resource type, target resource ID, IP address, and UTC timestamp.

- **Tracked Mutation Actions:** USER_LOGIN, REQUISITION_CREATE, SCORECARD_APPROVE, BULK_RESUME_UPLOAD, CANDIDATE_SCORE_EXECUTE, CLIENT_SUBMISSION, CLIENT_SHORTLIST, CLIENT_REJECT, INTERVIEW_PROPOSE, PIPELINE_TRANSITION, DPDP_ERASURE_EXECUTE.
- **Immutable Storage:** Audit log records have no update or delete API endpoints. They remain permanently queryable via GET /api/audit/logs by authorized recruiters to satisfy compliance audits.

# 16. UI/UX Design System & Architectural Decisions

HireAI's frontend interface was crafted following modern enterprise SaaS aesthetics inspired by Linear, Stripe Dashboard, and Vercel. The visual language balances high informational density with calm, distraction-free typography and consistent visual cues.

- **Color Palette:** Primary Background: Clean white (#ffffff) and slate-50 (#f8fafc). Primary Text: Deep navy (#0f172a). Primary Accent: Royal blue (#2563eb). Positive Actions (Shortlist, Approve): Emerald green (#10b981). Negative/Critical Actions (Reject, Erasure): Rose red (#ef4444). Neutral Borders: Slate-200 (#e2e8f0).
- **Typography:** Inter / Segoe UI sans-serif font family. Font scale strictly bounded from 10px (micro badges) to 24px (page headers). No oversized display headings.
- **Component Modularization:** Reusable component library in frontend/src/components/ui/: Button, Card, Badge, Modal, EmptyState, LoadingSkeleton. Reusable layout shell in components/layout/: AppShell, Sidebar, TopNav.
- **Responsive Viewport Optimization:** Desktop-optimized for Recruiters (multi-column tables, drawer sidebars, Kanban pipelines); mobile-responsive card layout for Client Hiring Managers reviewing shortlists on the go.

- **State Resilience:** Every screen implements four explicit states: (1) Loading skeletons with pulse animations, (2) Empty states with descriptive call-to-action buttons, (3) Error state alerts with retry buttons, and (4) Success toast notifications.

# 17. API Architecture & Endpoint Specification

HireAI provides a fully documented RESTful API organized into 11 modular routers under the /api prefix. All request and response payloads are validated using Pydantic v2 schemas.

| API Endpoint & HTTP Method | Functional Specification & RBAC Scope |
|---|---|
| POST /api/auth/login | Authenticates user via email/password; returns JWT bearer token and role. |
| GET /api/auth/me | Returns current authenticated user identity and organization context. |
| GET /api/dashboard/stats | Aggregates recruiter KPI metrics: active jobs, total candidates, weekly chart data. |
| GET / POST /api/requisitions | Lists requisitions or creates a new job requisition with client assignment. |
| POST /api/requisitions/{id}/analyze-jd | Calls Groq LLaMA 3.3 to extract structured scoring criteria from raw JD text. |
| PUT /api/requisitions/{id}/scorecard | Saves recruiter-edited and approved job scorecard criteria and weights. |
| POST /api/requisitions/{id}/upload | Ingests batch resumes (PDF, DOCX, XLSX, CSV); performs parsing and deduplication. |
| GET /api/upload-batches/template/excel | Downloads standardized Excel candidate bulk upload template (.xlsx). |
| GET /api/requisitions/{id}/candidates | Returns candidate list with search, status filters, and sorting parameters. |
| POST /api/requisitions/{id}/score-all | Executes deterministic 0–100 scoring engine across all requisition candidates. |
| GET /api/candidates/{id}/score | Retrieves candidate score breakdown and verbatim supporting CV evidence quotes. |
| POST /api/client/submissions | Submits candidate profile to client hiring manager with recruiter notes. |
| GET /api/client/submissions | Returns isolated candidate submissions for authenticated client tenant with masked PII. |

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|---|---|
| POST /api/client/submissions/{id}/shortlist | Client action to shortlist candidate and submit review comments. |
| POST /api/client/submissions/{id}/reject | Client action to reject candidate with mandatory rejection reason. |
| POST /api/client/submissions/{id}/interview | Client action to propose interview date and time. |
| GET / PATCH /api/requisitions/{id}/pipeline | Returns 11-stage pipeline distribution or updates candidate stage. |
| POST /api/candidates/{id}/deletion-request | Executes DPDP Right-to-Erasure: anonymizes database PII and purges CV files. |
| GET /api/audit/logs | Returns immutable security audit trail with user actions and IP addresses. |

## 18. Testing Strategy & Actual Test Results

HireAI employs a rigorous automated testing strategy utilizing pytest and httpx TestClient. Tests execute against an isolated database environment, validating authentication, RBAC boundaries, duplicate detection, document parsers, scoring arithmetic, client isolation, and DPDP workflows.

> **[AUTOMATED TEST EXECUTION VERIFICATION]** Actual Test Execution Outcome: 13 Tests Collected • 13 Passed • 0 Failed • Total Execution Time: 3.10 seconds. Test Platform: Windows (win32), Python 3.13.5, pytest 9.2.0, anyio 4.16.0.

| Test Case Identifier | Module Tested | Outcome | Test Verification Objective |
|---|---|---|---|
| test_login_success | auth.py | PASSED | Verifies JWT token generation and role identification on valid credentials. |
| test_login_invalid_credentials | auth.py | PASSED | Asserts 401 Unauthorized response on invalid email or wrong password. |
| test_recruiter_can_access_requisitions | rbac.py | PASSED | Verifies that recruiter role can read and create job requisitions. |
| test_client_cannot_create_requisition | rbac.py | PASSED | Asserts 403 Forbidden when client manager attempts recruiter operations. |

| | | | |
|---|---|---|---|
| test_duplicate_detection_by_hash_and_email | duplicate_detector.py | PASSED | Validates SHA-256 hash deduplication and normalized contact matching. |
| test_resume_parser_docx_and_cleaning | resume_parser.py | PASSED | Verifies python-docx extraction and text sanitization routines. |
| test_jd_analysis_mocked | groq_service.py | PASSED | Tests Groq AI extraction logic and deterministic offline fallback routines. |
| test_deterministic_scoring_fairness_and_evidence | scoring_engine.py | PASSED | Validates 0–100 scoring arithmetic, demographic exclusion, and CV quotations. |
| test_client_shortlist_and_rejection | client_portal.py | PASSED | Tests client shortlist, mandatory rejection reason, and interview proposals. |
| test_pipeline_transition | pipeline.py | PASSED | Verifies candidate lifecycle stage transitions and history audit logging. |
| test_candidate_deletion_request | privacy.py | PASSED | Validates DPDP Act Right-to-Erasure anonymization and file purge. |
| test_excel_template_download | resumes.py | PASSED | Verifies standard downloadable candidate Excel spreadsheet template generation. |
| test_excel_bulk_upload | excel_parser.py | PASSED | Validates batch spreadsheet ingestion of candidate records into PostgreSQL. |

# 19. Security Considerations & Threat Modeling

Security was integrated into HireAI's architecture at every development tier:

- **Password Security:** User passwords are encrypted with bcrypt using 12 salt rounds. Raw passwords are never logged, stored, or exposed in API responses.
- **Stateless Token Security:** Authentication relies on JWT tokens signed with HMAC-SHA256 (HS256) and an encrypted secret key. Tokens carry an expiration timestamp (default 24 hours).
- **Client Tenant Isolation:** Client queries strictly filter by client_organization_id derived from the verified JWT token, preventing cross-tenant data leakage.
- **Contact PII Masking:** Client-facing serialization schemas automatically mask email addresses and phone numbers before payload transmission.

- **File Upload Sanitization:** Uploaded filenames are sanitized via UUID prefixes. Files are validated against magic byte signatures to prevent malicious executable execution.

## 20. Operational Limitations

In the spirit of technical honesty and precision, the following current limitations of HireAI are documented:

- **OCR Processing Latency:** Tesseract OCR fallback on scanned PDF images is CPU-bound. Large batches of scanned image resumes may process at 2 to 4 seconds per page on standard machines without dedicated GPU acceleration.
- **Local Disk Storage Default:** In the current base deployment configuration, parsed resumes are stored in the local file system (./uploads) rather than an AWS S3 or Azure Blob container.
- **Single-Currency CTC Comparison:** The salary matching logic assumes candidate CTC and requisition budget are in the same currency (e.g. INR Lakhs or USD Thousands); automated real-time foreign currency conversion is not yet implemented.
- **Synchronous Batch Processing:** While FastAPI is asynchronous, extremely large file batches (e.g. 500 multi-page PDFs) benefit from an external task worker queue (e.g. Celery + Redis), which is slated for future architecture iterations.

## 21. Future Improvements & Roadmap

- **Distributed Worker Queue:** Integrate Celery / Redis or AWS SQS for asynchronous background processing of massive resume batches with webhooks.
- **Cloud Object Storage Integration:** Migrate file storage to AWS S3 / Cloudflare R2 with pre-signed URLs and server-side encryption (SSE-S3).
- **Calendar System Integration:** Connect Google Calendar and Microsoft Outlook APIs for real-time interviewer availability checking and automated calendar invitation dispatch.
- **Multi-Lingual Resume Parsing:** Incorporate multi-language NLP models to parse resumes submitted in non-English languages.
- **ATS Webhook Integrations:** Provide bidirectional webhook synchronizations with enterprise Applicant Tracking Systems (Greenhouse, Lever, Workday).

## 22. Operational & Design Assumptions

> **[EXPLICIT ASSUMPTIONS DECLARATION]** All assumptions made during the design, architecture, and implementation of HireAI are explicitly documented below. These assumptions define the technical operating boundaries of the application.

- **Assumption A-1 (Operating Environment):** Assumes the host operating system provides Python 3.11+ and Node.js 18+ with standard network connectivity to Groq Cloud API endpoints.
- **Assumption A-2 (Resume Formatting):** Assumes resumes are structured in standard English professional layouts (contact info header, education, skills, work experience history). Highly unconventional non-linear graphic designer portfolios may yield lower text extraction density.

- **Assumption A-3 (Scoring Primacy):** Assumes recruiters prioritize technical skill match and direct experience over secondary factors. Scoring weights remain configurable to override this assumption.
- **Assumption A-4 (Client Review Workflow):** Assumes client hiring managers act in good faith and provide genuine rejection reasons when declining submitted candidate profiles.
- **Assumption A-5 (Statutory Guidance):** Assumes data processing aligns with general DPDP Act 2023 principles (purpose limitation, consent logging, right-to-erasure); production enterprise deployment would undergo formalized legal counsel review.